

# Script-Controlled ACL - Restrict Record Access Based on Field Value

ServiceNow Micro-Project

PHASE 8 | PROJECT DEMONSTRATION PHASE

## 8. Project Demonstration Phase

This phase provides a structured walkthrough for presenting the completed project to reviewers or stakeholders, tying every configuration step back to a visible, verifiable output screen.

### 8.1 Demonstration Agenda

| # | Segment                                                  | Duration |
|---|----------------------------------------------------------|----------|
| 1 | Problem statement & objective recap                      | 2 min    |
| 2 | Walkthrough of users, roles, and table configuration     | 4 min    |
| 3 | Walkthrough of the four script-controlled ACLs           | 5 min    |
| 4 | Live impersonation demo (Read → Create → Write → Delete) | 6 min    |
| 5 | Q&A                                                      | 3 min    |

### 8.2 Demo Script — Step by Step

- Set the context:** "We have an Institution Details table shared by ECE, EEE, and CSE branches. Only EEE-branch users should see EEE rows; admins see everything."
- Show the foundation:** Open the EEE User record, the four custom roles (bb1–bb4), and the Institution Details table/fields.
- Show the security layer:** Open the four Access Control records and read out the Requires role, Data Condition, and Advanced script for each.
- Impersonate EEE User (bb1 only):** Open the list and show only the two EEE records appear.
- Remove bb1 (or impersonate a role-less user):** Refresh the list and show "No records to display".
- Impersonate Administrator:** Refresh the list and show every record across all three branches.
- Re-impersonate EEE User with bb1+bb2:** Show the New button appears, proving Create access is now enabled.
- Add bb3:** Edit the Description field of an EEE record in place and save, proving Write access.
- Add bb4:** Select a record → Actions on selected rows → Delete → confirm, proving Delete access.
- Close with the outcome statement** (Section 8.5).

### 8.3 Key Output Screens Presented During the Demo

| institution Details      |                     |        |             |                             |              |              |                 |
|--------------------------|---------------------|--------|-------------|-----------------------------|--------------|--------------|-----------------|
| Student Roll Numbe       |                     | Search |             | Actions on selected rows... |              |              |                 |
| All                      |                     |        |             |                             |              |              |                 |
| <input type="checkbox"/> | Student Roll Number | Branch | Description | Email                       | Faculty name | Phone Number | Student Name    |
|                          | INS0001007          | EEE    |             | saiguna@gmail.com           | Abel Tuter   | 9090898765   | Guna Sai        |
|                          | INS0001004          | EEE    |             | prashanth@gmail.com         | Abel Tuter   | 9090909090   | Prashanth Reddy |

Demo screen 1 — EEE User (bb1) sees only EEE-branch records.

| institution Details      |                     |                    |                  |                             |                     |              |
|--------------------------|---------------------|--------------------|------------------|-----------------------------|---------------------|--------------|
| Student Roll Numbe       |                     | Search             |                  | Actions on selected rows... |                     |              |
| All                      |                     |                    |                  |                             |                     |              |
| <input type="checkbox"/> | Student Roll Number | Faculty name       | Student Name     | Branch                      | Email               | Phone Number |
|                          | Search              | Search             | Search           | Search                      | Search              | Search       |
|                          | INS0001006          | ITIL User          | Praneeth         | ECE                         | praneeth@gmail.com  | 8989090906   |
|                          | INS0001008          | Malla Sravan kumar | Abraham Lincoln  | CSE                         | sravan@gmail.com    | 8978908989   |
|                          | INS0001005          | ITIL User          | Srinu Nandipalli | ECE                         | srinu@gmail.com     | 9089909090   |
|                          | INS0001007          | Abel Tuter         | Guna Sai         | EEE                         | saiguna@gmail.com   | 9090898765   |
|                          | INS0001001          | Abraham Lincoln    | Yaswanth Nikku   | CSE                         | yaswanth@gmail.com  | 8989898989   |
|                          | INS0001004          | Abel Tuter         | Prashanth Reddy  | EEE                         | prashanth@gmail.com | 9090909090   |

Demo screen 2 — Administrator sees all records across all branches.

institution Details

Student Roll Numbe

Search

Actions on selected rows...

New

All

Student Roll Number

Branch

Description

Email

Faculty name

Phone Number

Student Name

INS0001007

EEE

saiguna@gmail.com

Abel Tuter

9090898765

Guna Sai

INS0001004

EEE

prashanth@gmail.com

Abel Tuter

9090909090

Prashanth Reddy

Demo screen 3 — bb1 + bb2 roles unlock the New (Create) button for EEE records.

institution Details

Student Roll Numbe ▾

Search

Actions on selected rows... ▾

New

All

| <input type="checkbox"/> | Student Roll Number | Branch | Description     | Email               | Faculty name | Phone Number | Student Name    |
|--------------------------|---------------------|--------|-----------------|---------------------|--------------|--------------|-----------------|
| <input type="checkbox"/> | INS0001007          | EEE    | ACL is Executed | saiguna@gmail.com   | Abel Tuter   | 9090898765   | Guna Sai        |
| <input type="checkbox"/> | INS0001004          | EEE    |                 | prashanth@gmail.com | Abel Tuter   | 9090909090   | Prashanth Reddy |

Demo screen 4 — bb3 role enables editing (Write) of the EEE record's Description field.

institution Details

Student Roll Numbe

Search

Actions on selected rows...

New

All

Student Roll Number

Branch

Description

Email

Faculty name

Phone Num

INS0001007

EEE

ACL is Executed

saiguna@gmail.com

Abel Tuter

90908987

INS0001004

EEE

ACL is Executed

prashanth@gmail.com

Abel Tuter

90909090

Actions on selected rows...

Delete

Delete with preview...

Create Application File

Assign Tag:

New tag

Now Platform Security

Strategic Portfolio Management

Field Service Management

Now Platform

Security

IT Asset Management

IT Service Management

IT Operations Management

More...

Demo screen 5 — bb4 role enables Delete via the Actions on selected rows menu.

## 8.4 Anticipated Questions & Answers

| Question                                                       | Answer                                                                                                                                                                                |
|----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Why use a script instead of just a Data Condition?             | A Data Condition alone cannot special-case administrators or combine role logic cleanly; the script gives an explicit, auditable allow/deny decision.                                 |
| What stops a user from bypassing the ACL through the REST API? | ACLs are enforced by the platform's security engine at the data layer, so REST, Import Sets, and Service Portal all pass through the same rules as the UI.                            |
| Why four separate roles instead of one?                        | Separating bb1-bb4 by operation lets access be granted incrementally and audited per capability (read vs. create vs. write vs. delete), matching real-world least-privilege practice. |
| Does the admin role need to be added to "Requires role" too?   | No — the script checks <code>gs.hasRole('admin')</code> directly, so admins never need the custom bb roles to gain access.                                                            |

## 8.5 Closing / Outcome Statement

"By completing this lab, we've shown how script-controlled READ, WRITE, CREATE, and DELETE ACLs enforce record-level security, how user roles and record field values are evaluated together to make access decisions, and how ACLs protect sensitive data from unauthorized viewing, modification, creation, and deletion — across forms, lists, and Service Portal interfaces."

## 8.6 Reference Session

A recorded walkthrough session related to this ServiceNow project build is available here: [NM ServiceNow Project Development Session](#) (SmartBridge\_B, streamed live). Note: this document's screenshots were captured directly from the completed instance build (Section 8.3) rather than extracted from the video stream, since video frames cannot be programmatically captured from a live-stream page.